



Protocol for the Examination of Resection Specimens from Patients with Phyllodes Tumor of the Breast

Version: 1.2.0.0

Protocol Posting Date: September 2026

CAP Laboratory Accreditation Program Protocol Required Use Date: June 2027

The changes included in this current protocol version affect accreditation requirements. The new deadline for implementing this protocol version is reflected in the above accreditation date.

This protocol may be used for the following procedures AND tumor types:

Procedure	Description
Resection	Includes excision, segmental resection, lumpectomy, quadrantectomy, and partial or total mastectomy
Tumor Type	Description
Phyllodes tumor	

The following should NOT be reported using this protocol:

Procedure
Biopsy
Cytologic specimens

Important Note

The American Joint Committee on Cancer (AJCC) eighth edition and the World Health Organization (WHO) recommend staging malignant phyllodes tumors according to guidelines established for soft tissue sarcomas – extremity and trunk. T category, N category and stage group assignments do not apply to benign or borderline tumors. An abbreviated stage group table that only applies to malignant phyllodes tumors is included in the Explanatory Notes.

Version Contributors

Author(s): Stuart Schnitt, MD*, Kimberly Allison, MD, FCAP*, Uma Krishnamurti, MD, PhD, FCAP*, Hannah L Gilmore, MD, FCAP*, Brian Finkelman, MD, PhD, FCAP, Veronica Klepeis, MD, PhD, FCAP, Laura Rosenberger, MD, Puay Hoon Tan, MD

** Denotes primary author*

For any questions or comments, contact: cancerprotocols@cap.org.

Glossary:

Author: Expert who is a current member of the Cancer Committee, or an expert designated by the chair of the Cancer Committee.

Expert Contributors: Includes members of other CAP committees or external subject matter experts who contribute to the current version of the protocol.

Accreditation Requirements

Synoptic reporting with core and conditional data elements for designated specimen types* is required for accreditation.

- Data elements designated as core must be reported.
- Data elements designated as conditional only need to be reported if applicable.
- Data elements designated as optional are identified with "+". Although not required for accreditation, they may be considered for reporting.

This protocol is not required for recurrent or metastatic tumors resected at a different time than the primary tumor. This protocol is also not required for pathology reviews performed at a second institution (i.e., second opinion and referrals to another institution).

Full accreditation requirements can be found on the CAP website under [Accreditation Checklists](#). A list of core and conditional data elements can be found in the Summary of Required Elements under Resources on the CAP Cancer Protocols [website](#).

**Includes definitive primary cancer resection and pediatric biopsy tumor types.*

Synoptic Reporting

All core and conditionally required data elements outlined on the surgical case summary from this cancer protocol must be displayed in synoptic report format. Synoptic format is defined as:

- Data element: followed by its answer (response), outline format without the paired Data element: Response format is NOT considered synoptic.
- The data element should be represented in the report as it is listed in the case summary. The response for any data element may be modified from those listed in the case summary, including "Cannot be determined" if appropriate.
- Each diagnostic parameter pair (Data element: Response) is listed on a separate line or in a tabular format to achieve visual separation. The following exceptions are allowed to be listed on one line:
 - Anatomic site or specimen, laterality, and procedure
 - Pathologic Stage Classification (pTNM) elements
 - Negative margins, as long as all negative margins are specifically enumerated where applicable
- The synoptic portion of the report can appear in the diagnosis section of the pathology report, at the end of the report or in a separate section, but all Data element: Responses must be listed together in one location
- Organizations and pathologists may choose to list the required elements in any order, use additional methods in order to enhance or achieve visual separation, or add optional items within the synoptic report. The report may have required elements in a summary format elsewhere in the report IN ADDITION TO but not as replacement for the synoptic report i.e., all required elements must be in the synoptic portion of the report in the format defined above.

Summary of Changes

v 1.2.0.0

- Protocol now required for accreditation for malignant phyllodes
- WHO 6th Edition updates to content and explanatory notes
- Update to "Tumor Site" question
- Updates to instructional Display Items (Histologic Type, Stromal Overgrowth, Mitotic Rate, Histologic Tumor Border, and Margin Status)
- Addition of optional "Tumor Necrosis" question
- Updates to MARGINS and ADDITIONAL FINDINGS sections
- pTNM Classification section updates to include the addition of Modified Classification and T Suffix
- eCP only ASCII technical update

Reporting Template

Protocol Posting Date: September 2026

Select a single response unless otherwise indicated.

CASE SUMMARY: (PHYLLODES OF THE BREAST: Resection)

Standard(s): AJCC 8

Note: Use of this reporting template is required for accreditation purposes for malignant phyllodes tumors, but is optional for benign and borderline phyllodes. The template can be used for benign and borderline phyllodes tumors, but pathologic stage classification should only be done for those tumors classified as malignant.

SPECIMEN

Procedure

- ☐ Excision (less than total mastectomy, including lumpectomy and partial mastectomy)
☐ Total mastectomy (including nipple-sparing and skin-sparing mastectomy)
☐ Other (specify): _____
☐ Not specified

Specimen Laterality

- ☐ Right
☐ Left
☐ Not specified

TUMOR

+Tumor Site

Tumor Site descriptor should specify the location of the invasive cancer based on correlation with radiology designation and / or gross findings (e.g., "R1, 3:00, 2 cm from nipple" or "upper outer quadrant").

- ☐ Specify tumor site / location: _____
☐ Not specified

Tumor Size

- ☐ Greatest dimension in Millimeters (mm): _____ mm
+Additional Dimension in Millimeters (mm): ____ x ____ mm
☐ Cannot be determined (explain): _____

Histologic Type (Note [A](#))

Per the 6th edition of the WHO Breast Tumours a diagnosis of malignant phyllodes tumor requires the presence of at least four of these five adverse features: marked stromal cellularity, marked stromal atypia, stromal overgrowth, an infiltrative tumor border and greater than or equal to 10 mitoses per 10 high-power fields (HPFs). Malignant phyllodes tumor is also diagnosed when malignant heterologous elements other than pure well-differentiated liposarcoma are present, even if not all of the other histologic features of malignancy are observed. Tumors with up to three of these adverse features should be classified as borderline when some but not all of these changes are present. See Note A for additional details.

- ☐ Phyllodes tumor, benign
☐ Phyllodes tumor, borderline
☐ Phyllodes tumor, malignant

+Histologic Type Comment: _____

Stromal Cellularity (Note [B](#))

- ☐ Mild (stromal nuclei are non-overlapping)
- ☐ Moderate (some overlapping stromal nuclei)
- ☐ Marked (many overlapping stromal nuclei)
- ☐ Cannot be determined (explain): _____

Stromal Atypia (Note C)

- ☐ None
- ☐ Mild (minimal variation in nuclear size, even chromatin, and smooth nuclear contours)
- ☐ Moderate (more variation in nuclear size and irregular nuclear membranes)
- ☐ Marked (marked nuclear pleomorphism, hyperchromasia, and irregular nuclear contours)
- ☐ Cannot be determined (explain): _____

Stromal Overgrowth (Note D)

Stromal overgrowth is present when there is at least one low-power microscopic field (4x objective and 10x eyepiece or 22.9 mm²) that contains stroma only without epithelial elements. Expansion of the stroma due to edema, infarction, and myxoid change should not be considered in assessing stromal overgrowth.

- ☐ Absent
- ☐ Present
- ☐ Cannot be determined (explain): _____

Mitotic Rate (Note E)

Malignant phyllodes tumors typically have greater than or equal to 10 mitoses per 10 high-power fields (40x objective and 10x eyepiece) or greater than or equal to 5 mitoses / mm². Benign phyllodes tumors typically have less than 5 mitoses per 10 HPFs (less than 2.5 mitoses / mm²). Use the most mitotically active area for mitotic rate, which is often in the most cellular areas of stroma.

- ☐ None identified: _____
- ☐ Specify number of mitoses per 10 high power fields: _____ mitoses per 10 High Power Fields (HPFs)
- ☐ Specify number of mitoses per square Millimeter (mm): _____ mitoses per mm²

Histologic Tumor Border

A circumscribed border is smooth and well-defined or shows a minimally irregular tumor interface with adjacent stroma. Infiltrative (permeative) tumor borders can be focally infiltrative (unequivocal invasion into adjacent stroma in one low-power field) and / or limited (not extending very deep into peripheral benign tissues) or extensively infiltrative (unequivocal invasion in a wide area or in multiple foci along the tumor periphery).

- ☐ Circumscribed (well-defined; pushing)
- ☐ Infiltrative (permeative)
 - + ☐ Focal / limited
 - + ☐ Extensive
- ☐ Cannot be determined (explain): _____

Malignant Heterologous Elements (Note F)

A phyllodes tumor is regarded as malignant when there are malignant heterologous elements, even when not all of the other histological features of malignancy are present. This rule does not apply if the only heterologous element is well-differentiated liposarcoma. In the breast, those tumors usually lack MDM2 and CDK4 amplifications and have a low metastatic risk. A diagnosis of malignant phyllodes tumor should therefore not be based solely on the presence of well-differentiated liposarcoma without other histologic features of malignancy.

- ☐ Not identified
- ☐ Liposarcoma (excluding well-differentiated liposarcoma)

CAP
Approved

Breast.Phyllodes_1.2.0.0.REL_CAPCP

___ Osteosarcoma
___ Chondrosarcoma
___ Other (specify): _____

+Tumor Necrosis (Note [G](#))

___ Not identified
___ Present

+Necrosis Comment: _____

MARGINS

Final Margin Status for Phyllodes Tumor (Note [H](#))

Final margin status should incorporate any additional margins provided for an overall final margin status. Margin status is listed as positive if there is ink on phyllodes tumor (i.e., the distance is 0 mm). For benign phyllodes tumors, additional details about the distance to closest negative margins are not required (but can be provided either in detail or more limited detail using the "Other (specify)" option). For borderline and malignant phyllodes tumors, additional detail about the distance to margins out to 10 mm should ideally be provided.

___ All final margins greater than 10 mm from Phyllodes tumor
___ Phyllodes tumor present within 0 to 10 mm of final margins

Margin(s) Involved by Phyllodes Tumor (at ink)

___ None identified
___ Specify involved margin(s): _____

Margin(s) Less than 1 mm from Phyllodes Tumor (but not at ink)

___ None identified
___ Specify margin(s): _____

Margin(s) 1 to 2 mm from Phyllodes Tumor

___ None identified
___ Specify margin(s) and distance(s): _____

Margin(s) Greater than 2 mm but Less than or Equal to 5 mm from Phyllodes Tumor

___ None identified
___ Specify margin(s) and distance(s): _____

Margin(s) Greater than 5 mm but Less than or Equal to 10 mm from Phyllodes Tumor

___ None identified
___ Specify margin(s) and distance(s): _____

+Margin(s) Greater than 10 mm from Phyllodes Tumor

___ None identified
___ Specify margin(s) and distance(s): _____
___ Other (specify): _____
___ Cannot be determined (explain): _____

+Margin Comment: _____

REGIONAL LYMPH NODES

Regional Lymph Node Status

___ Not applicable (no regional lymph nodes submitted or found)
___ Regional lymph nodes present

- ☐ All regional lymph nodes negative for tumor
- ☐ Tumor present in regional lymph nodes
- Number of Lymph Nodes with Tumor**
- ☐ Exact number (specify): _____
- ☐ At least (specify): _____
- ☐ Other (specify): _____
- ☐ Cannot be determined (explain): _____
- ☐ Other (specify): _____
- ☐ Cannot be determined (explain): _____
- Number of Lymph Nodes Examined**
- ☐ Exact number (specify): _____
- ☐ At least (specify): _____
- ☐ Other (specify): _____
- ☐ Cannot be determined (explain): _____

+Regional Lymph Node Comment: _____

DISTANT METASTASIS

Distant Site(s) Involved, if applicable

- ☐ Not applicable
- ☐ Other (specify): _____
- ☐ Cannot be determined (explain): _____

pTNM CLASSIFICATION (AJCC 8th Edition) (Note 1)

Staging applies only to malignant phyllodes tumors. pT and pN categories should not be assigned for benign and borderline tumors.

pTNM Classification (AJCC 8th Edition) (required only if the tumor is malignant)

Reporting of pT, pN, and (when applicable) pM categories is based on information available to the pathologist at the time the report is issued. As per the AJCC (Chapter 1, 8th Ed.) it is the managing physician's responsibility to establish the final pathologic stage based upon all pertinent information, including but potentially not limited to this pathology report.

- ☐ Not applicable (tumor is not graded as malignant)
- ☐ Tumor is malignant

The following section applies only if the tumor is malignant. Do not assign pT and pN stage categories for benign or borderline tumors.

Modified Classification (required only if applicable) (select all that apply)

- ☐ Not applicable
- ☐ y (post-neoadjuvant therapy)
- ☐ r (recurrence)

pT Category

- ☐ pT not assigned (cannot be determined based on available pathological information)
- ☐ pT0: No evidence of primary tumor
- ☐ pT1: Tumor 5 cm or less in greatest dimension
- ☐ pT2: Tumor more than 5 cm but not more than 10 cm
- ☐ pT3: Tumor more than 10 cm but not more than 15 cm
- ☐ pT4: Tumor more than 15 cm in greatest dimension

T Suffix (required only if applicable)

- ☐ Not applicable
☐ (m) multiple primary synchronous tumors in a single organ

pN Category

When no lymph nodes are present, the pathologic 'N' category is not assigned (pNX is not used and should not be reported)

- ☐ pN not assigned (no nodes submitted or found)
☐ pN not assigned (cannot be determined based on available pathological information)
☐ pN0: No regional lymph node metastasis
☐ pN1: Regional lymph node metastasis

pM Category (required only if confirmed pathologically)

- ☐ Not applicable - pM cannot be determined from the submitted specimen(s)
☐ pM1: Distant metastasis

ADDITIONAL FINDINGS

+Additional Lesion(s) (select all that apply)

If there is co-existing invasive carcinoma or DCIS, consider using the additional corresponding CAP template.

Subtypes of LCIS include: Pleomorphic LCIS (pleomorphic nuclei greater than 4 times the size of a lymphocyte or equivalent to nuclei of high-grade DCIS) and Florid LCIS (proliferation of cells cytologically similar to those of classic LCIS but expanding the acini of the involved TDLUs so that little to no residual intervening intra-lobular stroma is present, and / or an expanded acinus or duct spans approximately 40-50 cells in diameter). Comedonecrosis in classic LCIS may also be considered variant (describe in other).

- ☐ Not identified
☐ Fibroepithelial proliferation (coexisting fibroadenoma or fibroadenomatoid change in the tissue surrounding the phyllodes tumor)
☐ Lobular carcinoma in situ, classic
☐ Lobular carcinoma in situ, pleomorphic
☐ Lobular carcinoma in situ (specify): _____
☐ Atypical lobular hyperplasia
☐ Atypical ductal hyperplasia
☐ Flat epithelial atypia
☐ Other (specify): _____

+Additional Lesion(s) Comment: _____

COMMENTS

Comment(s): _____

Explanatory Notes

A. Histologic Type/Grade

Phyllodes tumors are classified into three types: benign, borderline or malignant (also referred to as grades) that are associated with different local and distant recurrence rates.^{1,2,3,4,5} Six histologic features are evaluated in the process of assigning the histologic type/grade as indicated in Table 1.

Per the 6th edition of the WHO Tumours of the Breast, a malignant phyllodes tumor is diagnosed when four or more of the five most adverse/malignant histological features are present: marked stromal hypercellularity; marked stromal atypia; stromal overgrowth; an infiltrative (permeative) tumor border; and greater than or equal to 10 mitotic figures in 10 high-power fields. A phyllodes tumor is also categorized as malignant if there is a malignant heterologous mesenchymal component (e.g., liposarcoma, chondrosarcoma, osteosarcoma) even if the other histological parameters are not present, or if only some are present. An exception to this rule is if the heterologous element is an atypical lipomatous tumor/well-differentiated liposarcoma. Well-differentiated liposarcomas in the breast usually lack MDM2 and CDK4 amplifications and appear to have a low metastatic risk. Hence, a diagnosis of malignant phyllodes tumor should not be based solely on the presence of well-differentiated liposarcoma without the other histologic features that support malignancy.

Tumors with up to three adverse/malignant histologic features or at least three of the borderline histologic features (and no malignant heterologous differentiation) can be classified as borderline. The combination of features present should be included in the report.

Benign phyllodes tumors typically have mild stromal hypercellularity, minimal to no stromal atypia, no stromal overgrowth, circumscribed (pushing) tumor borders, and less than 5 mitoses per 10 high-power fields (HPFs).

The WHO recognizes that tumors that have features that straddle different types/grades can be challenging to characterize and recommend comprehensive tissue sampling to account for potentially heterogeneous lesions (at least 2 sections per 10 mm). While some diagnostic approaches differentially weigh some histologic features to provide an overall grade, the WHO 6th edition recommends the following pragmatic approach: when grading tumors with features of both the benign and borderline grades, the pathologist can assign the diagnostic category that best reflects the predominant/majority of the features, accompanied by an explanatory comment. For example, when one or at most two borderline features are focally observed or present in a limited extent in an otherwise benign appearing phyllodes tumor, the tumor may still be classified as favoring benign, with documentation/comment about the higher risk features present. Similarly, when overlapping features between borderline and malignant are present, if features are predominantly borderline with one to three malignant features (without malignant heterologous differentiation), the case may be diagnosed as borderline with explanatory comments. Classification of such tumors may benefit from consultation with subspecialty breast pathologist colleagues.

Table 1. Histologic Features of Phyllodes Tumors

Histologic feature	Benign	Borderline	Malignant
Stromal cellularity	Mild	Moderate	Marked
Stromal atypia	Mild or none	Mild or moderate	Marked
Stromal overgrowth	Absent	Absent or very focal	Present
Mitotic rate	<5 mitoses per 10 HPFs or <2.5 mitoses per mm ²	5 - 9 mitoses per 10 HPFs or 2.5 - <5 mitoses/mm ²	≥10 mitoses per 10 HPFs or ≥5 mitoses/mm ²
Tumor border	Circumscribed	Usually circumscribed but may be focally infiltrative	Focally or extensively infiltrative (permeative)
Malignant heterologous stromal elements	Absent	Absent	Sometimes present

HPF: High power field (40x objective and 10x eyepiece)

References

1. WHO Classification of Tumours Editorial Board. *Breast tumours*. Lyon (France): International Agency for Research on Cancer; 2026. (WHO classification of tumours series, 6th ed., beta version).
2. Tan PH, et al. Malignant phyllodes tumours of the breast: the case for revising WHO's 'full house' diagnostic criteria. *Histopathology*. 2025 Aug;87(2):169-182.
3. Li X, et al. Validation Study of the Newly Proposed Refined Diagnostic Criteria for Malignant Phyllodes Tumor With 136 Borderline and Malignant Phyllodes Tumor Cases. *Am J Surg Pathol*. 2024 Sep 1;48(9):1146-1153.
4. Turashvili G, et al. Comprehensive Clinical-Pathologic Assessment of Malignant Phyllodes Tumors: Proposing Refined Diagnostic Criteria. *Am J Surg Pathol*. 2023 Nov 1;47(11):1195-1206.
5. Tan PH, Thike AA, Tan WJ, et al. Predicting clinical behaviour of breast phyllodes tumours: a nomogram based on histological criteria and surgical margins. *J Clin Pathol*. 2012; 65:69-76.

B. Stromal Cellularity

Mild hypercellularity is characterized by a slight increase in stromal cells as compared with normal perilobular stroma, with evenly spaced nuclei that are not touching or overlapping, while marked stromal cellularity shows confluent areas of densely overlapping nuclei. Moderate stromal cellularity has findings that are intermediate between the two, with some overlapping stromal nuclei.¹

References

1. WHO Classification of Tumours Editorial Board. *Breast tumours*. Lyon (France): International Agency for Research on Cancer; 2026. (WHO classification of tumours series, 6th ed., beta version).

C. Stromal Atypia

Mild stromal atypia is reported when there is little variation in nuclear size and the nuclear contours are smooth. Cases with moderate atypia show some variation in the size of stromal nuclei and some wrinkling of nuclear membranes. Marked stromal atypia is identified when there is marked variation in nuclear size, coarse chromatin and irregular nuclear membranes with discernible nucleoli.¹

References

1. WHO Classification of Tumours Editorial Board. *Breast tumours*. Lyon (France): International Agency for Research on Cancer; 2026. (WHO classification of tumours series, 6th ed., beta version).

D. Stromal Overgrowth

Stromal overgrowth is defined by the absence of epithelial elements in at least one low-power microscopic field containing only stroma. A low-power field can be defined either as a 4x objective and 10x eyepiece or as 22.9 mm². Expansion of the stroma due to edema, infarction, and myxoid change should not be considered in assessing stromal overgrowth.¹

References

1. WHO Classification of Tumours Editorial Board. *Breast tumours*. Lyon (France): International Agency for Research on Cancer; 2026. (WHO classification of tumours series, 6th ed., beta version).

E. Mitotic Rate

A diagnosis of malignant phyllodes tumor typically involves at least 10 mitoses per 10 high-power fields (40x objective and 10x eyepiece) or at least 5 mitoses/mm². Mitotic activity in benign phyllodes tumor is usually low (<5 mitoses/10HPF or less than 2.5 mitoses per mm²). Borderline phyllodes tumors usually have 5 to 9 mitoses per 10 HPF (2.5 to <5 mitoses/mm²). Mitotic figures given in per 10 HPF correspond to a HPF field that is 0.5 mm in field diameter and 0.2 mm² in area.¹

Use the most mitotically active area for mitotic rate, which is often in the most cellular areas of stroma.

To report the number of mitoses per square millimeter, the area of the high-power field must be known, but microscopes vary in field size, so the area must be determined for each microscope. The diameter of an HPF can be determined using a micrometer or calculated by using the method below:

Using a clear ruler, measure the diameter of a low-power field. This number can be used to calculate a constant based on the following formula:

Eyeiece Magnification x Objective Magnification x Microscopic Field Diameter = A Constant

Once the value of the constant is known, the diameter of the high-power field can be calculated by using the following formula:

High-Power Field Diameter = Constant / (Eyeiece Magnification x Objective Magnification)

Half of the field diameter is the radius of the field (r), which can then be used to calculate the area of the HPF:

Area of High-Power Field = r² x 3.1415

References

1. WHO Classification of Tumours Editorial Board. *Breast tumours*. Lyon (France): International Agency for Research on Cancer; 2026. (WHO classification of tumours series, 6th ed., beta version).

F. Malignant Heterologous Elements

Malignant heterologous elements include osteosarcoma, chondrosarcoma, rhabdomyosarcoma, and rarely other types of sarcoma. The presence of well differentiated liposarcoma alone is not used to categorize a phyllodes tumor as malignant.¹

References

1. WHO Classification of Tumours Editorial Board. *Breast tumours*. Lyon (France): International Agency for Research on Cancer; 2026. (WHO classification of tumours series, 6th ed., beta version).

G. Tumor Necrosis

Tumor necrosis is mostly associated with malignant phyllodes tumors and has been reported to be a poor prognostic factor.^{1,2} This must be distinguished from tumor infarction, which is generally a hemorrhagic necrosis.³

References

1. Chia-Yun Yu, Tsai-Wei Huang, Ka-Wai Tam. Management of phyllodes tumor: A systematic review and meta-analysis of real-world evidence. *Int J Surg*. 2022; 107:106969. PMID: 36328344. DOI: 10.1016/j.ijssu.2022.106969.
2. Hae Jin Park, Han Suk Ryu, Kyubo Kim, Kyung Hwan Shin, Wonshik Han, Dong-Young Noh. Risk Factors for Recurrence of Malignant Phyllodes Tumors of the Breast. *In Vivo*. 2019 Jan-Feb;33(1):263-269. PMID: 30587634 doi: 10.21873/invivo.11470.
3. N Tagaya, H Kodaira, H Kogure, K Shimizu: A Case of Phyllodes Tumor with Bloody Nipple Discharge in Juvenile Patient. *Breast Cancer*. 1999 Jul 25;6(3):207-210. PMID: 11091717; DOI: 10.1007/BF02967169.

H. Final Margin Status

The optimal margin width for phyllodes tumors is often based on a variety of factors, including histologic type. The National Comprehensive Cancer Network (NCCN) guidelines recommend complete mass removal for benign phyllodes tumors but without the intent of obtaining widely negative surgical margins.¹ Therefore, for benign phyllodes tumors, additional details about the specific distances to negative margins is not required (but can be provided). For borderline and malignant phyllodes tumors, the NCCN² recommends oncologically appropriate margins with the initial intent for > 1 cm margins when achievable and re-excision for positive margins. Therefore, additional detail about the distance to final margins out to 10 mm should ideally be provided.

Final margin status should incorporate any additional margins provided for an overall final margin status. Margin status is listed as positive if there is ink on phyllodes tumor (i.e., the distance is 0 mm).

References

1. National Comprehensive Cancer Network (NCCN). NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®): Breast Cancer, Special Consideration: Phyllodes Tumor. Version 6.2026. Plymouth Meeting, PA: National Comprehensive Cancer Network; 2026. <https://www.nccn.org/guidelines/guidelines-detail?category=1&id=1419> Accessed Sept 1, 2026.
2. National Comprehensive Cancer Network (NCCN). NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®): Soft Tissue Sarcoma. Version 5.2026. Borderline/Malignant Phyllodes Tumor of the Breast. Plymouth Meeting, PA: National Comprehensive Cancer Network; 2026. <https://www.nccn.org/guidelines/guidelines-detail?category=1&id=1464> Accessed Sept 1, 2026.

I. pTNM CLASSIFICATION

The American Joint Committee on Cancer (AJCC) eighth edition¹ and the World Health Organization (WHO)² recommend staging malignant phyllodes tumors according to guidelines established for soft tissue sarcomas – extremity and trunk. T category, N category, and stage group assignments do not apply to benign or borderline phyllodes tumors and should only be reported if the tumor is malignant.

AJCC Prognostic Stage Groups

T	N	M	Stage group
T1	N0	M0	II
T2	N0	M0	IIIA
T3, T4	N0	M0	IIIB
Any T	N1	M0	IV
Any T	Any N	M1	IV

TNM Descriptors

For identification of special cases of TNM or pTNM classifications, the “m” suffix and “y” and “r” prefixes are used. Although they do not affect the stage grouping, they indicate cases needing separate analysis.

The “m” suffix indicates the presence of multiple primary tumors in a single site and is recorded in parentheses: pT(m)NM.

The “y” prefix indicates those cases in which classification is performed during or after initial multimodality therapy (i.e., neoadjuvant chemotherapy, radiation therapy, or both chemotherapy and radiation therapy).

The cTNM or pTNM category is identified by a “y” prefix. The ycTNM or ypTNM categorizes the extent of tumor actually present at the time of that examination. The “y” categorization is not an estimate of tumor before multimodality therapy (i.e., before initiation of neoadjuvant therapy).

The “r” prefix indicates a recurrent tumor when staged after a documented disease-free interval and is identified by the “r” prefix: rTNM.

T Category Considerations

Only malignant phyllodes tumors are staged according to AJCC staging rules. The pathologic 'T' category (pT) is not assigned for benign and borderline phyllodes tumors.

N Category Considerations

Regional nodal metastasis is uncommon in phyllodes tumor and lymph nodes may not be sampled. When no lymph nodes are resected or present in the specimen, the pathologic 'N' category is not assigned; pNX should not be used.

References

1. Maki RG, Folpe AL, Guadagnolo BA, et al. Chapter 45. Soft tissue sarcoma – Unusual histologies and sites. In: Amin MB, ed. *AJCC Cancer Staging Manual*. 8th ed. New York: Springer; 2017:539-544.
2. WHO Classification of Tumours Editorial Board. *Breast tumours*. Lyon (France): International Agency for Research on Cancer; 2026. (WHO classification of tumours series, 6th ed., beta version).